🫁 Breathing, Exercise & Energy: Guided Notes · ANSWER KEY

Biology A · Unit 5 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A2 Exercise lab, A3 Combustion, A4 Yeast Feast, A5 Respiration + ATP, heart/lungs, A7

Name: _____

Use the class slides and your reading. Key words:

cellular respiration · *ATP* · *mitochondria* · *oxygen* · *carbon dioxide* · *combustion* · *yeast* · *BTB* · *heart rate*

A. Cellular respiration (A5)

1. Cellular respiration happens in the **mitochondria** .

2. Fill in the table.

Inputs	Outputs
glucose	**carbon dioxide**
oxygen	**water**
—	**ATP (energy)**

3. ATP releases energy when a **phosphate** group comes off.

B. Exercise lab (A2)

4. **Breathing rate change**

> Example: 14 breaths/min at rest, 28 after exercise → 28 − 14 = 14 more breaths per minute

a) Heart rate 70 at rest, 110 after exercise. Change = **40 beats per minute**

b) BTB turned yellow in 90 s at rest and 45 s after exercise. How many times faster? **2 times faster**

5. What do the lab results show about exercise and CO2?
After exercise the body makes more CO2 (BTB turned yellow faster) because cells do more respiration.

C. Combustion and yeast (A3, A4)

6. A burning marshmallow loses **mass** because atoms leave as CO2 and water.

7. In the Yeast Feast, which tube made the most CO2 and why?
Glucose: it is the simple sugar yeast can use right away.

🔑 *Key written by our team from Ms. Dunham's materials, please check.*